Lecture 5 - 20/05

Thursday, May 21, 2026 1:53 PM

ecological systems

Population and the rate of its change

$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \frac{\Delta x}{x} = \frac{\dot{x}}{x}$

if food is abundant $\frac{\dot{x}}{x} = c$, $c > 0$, population will grow
since there are no infinite food sources so, $\frac{\dot{x}}{x} = c - x$
this can be written as $\dot{x} = cx - x^2 = rx(1 - \frac{x}{K})$
this is one dim, but we can grow to see how species interact

$$\begin{cases} \frac{\dot{x}}{x} = f(x,y) \\ \frac{\dot{y}}{y} = g(x,y) \end{cases}$$

Lotka-Volterra systems

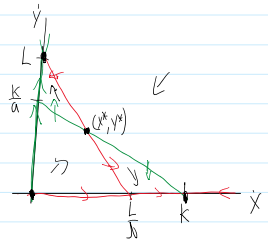
$$\begin{cases} \frac{\dot{x}}{x} = c_x + a_{xx}x + a_{xy}y \\ \frac{\dot{y}}{y} = c_y + a_{yx}x + a_{yy}y \end{cases}$$

competing populations

$$\begin{cases} \frac{\dot{x}}{x} = K - x - ay \\ \frac{\dot{y}}{y} = L - bx - y \end{cases}$$

K/L are the rate of growth just given the food while they represent the same constant food source they control the rate of growth given that food which changes per species

$$\begin{cases} \dot{x} = x(K - x - ay) \\ \dot{y} = y(L - bx - y) \end{cases}$$ Params: $K, L, a, b > 0$



looking for nullcline:

$0 = \dot{x} = x(K - x - ay)$ $\quad x = 0$ $\quad K - x - ay = 0$, $y = -\frac{1}{a}x + \frac{K}{a}$ vertical

$0 = \dot{y} = y(L - bx - y)$ $\quad y = 0$ $\quad L - bx - y = 0$, $y = -bx + L$ horizontal

FP - $(0,0), (K,0), (0,L), (x^*, y^*)$ $\quad x^*, y^* > 0$ (x^*, y^*) can be negative depending on params or not exist,

lets look at different cases

① $L > \frac{K}{a}$, $K > \frac{L}{b}$ $\quad \dot{\begin{pmatrix} x \\ y \end{pmatrix}} = F\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix}$

$$DF\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{\partial f(x,y)}{\partial x} & \frac{\partial f(x,y)}{\partial y} \\ \frac{\partial g(x,y)}{\partial x} & \frac{\partial g(x,y)}{\partial y} \end{pmatrix}_{FP}$$

$$DF\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (K - x - ay) - x & -ax \\ -by & (L - bx - y) - y \end{pmatrix}$$

reminder
$q(\lambda) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = (a-\lambda)(d-\lambda) - bc =$
$\lambda^2 - \lambda(a+d) + ad - bc$
$\lambda_1 \cdot \lambda_2 = \det M$
$\lambda_1 + \lambda_2 = \text{trace } M$
we can determine λ_1, λ_2 sign just with det and trace
if $\det M > 0$ and $\text{tr } M < 0$
$\lambda_1, \lambda_2 < 0$ and FP is stable

$$DF\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix} (k-\alpha x-ay)-x & -ax \\ -by & (L-bx-y)-y \end{pmatrix}$$

$$DF\begin{pmatrix}0\\0\end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & L \end{pmatrix} \rightarrow \text{unstable FP}$$

$$DF\begin{pmatrix}k\\0\end{pmatrix} = \begin{pmatrix} -k & -ak \\ 0 & L-bk \end{pmatrix} \quad \lambda_1\lambda_2 = -\cdot - = + > 0, \quad \lambda_1+\lambda_2 = -+- = - < 0 \rightarrow \text{stable FP}$$

$$DF\begin{pmatrix}0\\L\end{pmatrix} = \begin{pmatrix} k-aL & 0 \\ -bL & -L \end{pmatrix} \quad \det > 0, \ tr < 0 \rightarrow \text{stable}$$

$$DF\begin{pmatrix}x^*\\y^*\end{pmatrix} = \begin{pmatrix} -x^* & -ax^* \\ -by^* & -y^* \end{pmatrix} \quad \det = x^*y^* - abx^*y^* = x^*y^*(1-ab) < 0, \ tr < 0 \rightarrow \text{saddle FP}$$

$L > \frac{k}{a} > \frac{L}{ab}$

$1 > \frac{1}{ab}$

$ab > 1$

det and trace

if $\det M > 0$ and $tr\, M < 0$

$\lambda_1, \lambda_2 < 0$ and FP is stable

if $\det M < 0$

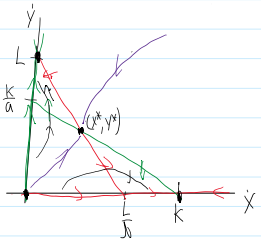
$\lambda_1 < 0, \lambda_2 > 0$ and FP is saddle point

these are two stable manifold W^s

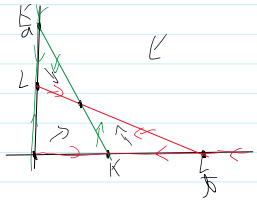
on these the populations can coexist

anything above W^s, will go to (0,L), meaning X population is extinct

anything below W^s, will go to (k,0), meaning X population is extinct



(2) $\frac{k}{a} > L$, $\frac{L}{b} > k$



(0,0) is still unstable

$$DF\begin{pmatrix}k\\0\end{pmatrix} = \begin{pmatrix} -k & -ak \\ 0 & L-bk \end{pmatrix} \quad \det < 0 \rightarrow \text{saddle}$$

($L-bk > 0$)

$$DF\begin{pmatrix}0\\L\end{pmatrix} = \begin{pmatrix} k-aL & 0 \\ -bL & -L \end{pmatrix} \quad \det < 0 \rightarrow \text{saddle}$$

($k-aL > 0$)

$$DF\begin{pmatrix}x^*\\y^*\end{pmatrix} = \begin{pmatrix} -x^* & -ax^* \\ -by^* & -y^* \end{pmatrix} \quad \det = x^*y^*(1-ab) > 0, \ tr < 0$$

$k < \frac{L}{b} < \frac{k}{ak} \rightarrow$ stable

$1 < \frac{1}{ab}$

$ab < 1$

So in this case, no matter the start as long as neither population is 0, they will co-exist

Predator-prey

prey $\dot{x} = a - by$

predator $\dot{y} = -c + ex$

a - prey's natural growth without predators

-c - predators natural death without prey

$a, b, c, e > 0$

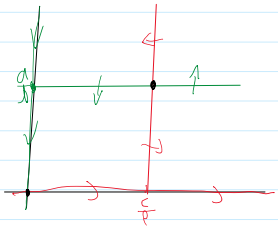
$$\begin{cases} \dot{x} = x(a-by) \\ \dot{y} = y(-c+ex) \end{cases}$$

nullclines

$\dot{x} = 0 \Rightarrow x(a-by) = 0 \Rightarrow x = 0$, $a - by = 0$

$\dot{y} = 0 \Rightarrow y(-c+ex) = 0 \Rightarrow y = 0$, $-c + ex = 0$

FP



there are 2 FP, (0,0) and $\left(\frac{c}{e}, \frac{a}{b}\right)$ (where green and red meet)

nullcline flow direction

$X=0 \Rightarrow \dot{Y} = -YC < 0$ so flows down

by continuity, $a-bY=0$ when $X < \frac{a}{b}$, goes down

than $X > \frac{a}{b} \Rightarrow eX > C \Rightarrow (-C+eX) > 0$, so $\dot{Y} > 0$

$Y=0 \Rightarrow \dot{X} = aX > 0$, so flows to the right

by cont $-C+eX=0$, when $Y < \frac{C}{b}$

when $Y > \frac{C}{b} \Rightarrow Yb > a \Rightarrow (a-bY) < 0$ so $\dot{X} < 0$

to understand how the FP acts, analyze λ

$$DF\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} a-bY & -Xb \\ eY & -C+eX \end{pmatrix}$$

$$DF\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & -c \end{pmatrix} \quad \det < 0 \Rightarrow \text{saddle}$$

$$DF\begin{pmatrix} \frac{c}{e} \\ \frac{a}{b} \end{pmatrix} = \begin{pmatrix} 0 & -\frac{bc}{e} \\ \frac{ae}{b} & 0 \end{pmatrix} \quad \varphi(\lambda) = \lambda^2 + ac \quad \lambda_{1,2} = \pm i\sqrt{ac}$$

pure imaginary λ, this is a cycle

$\frac{\dot{Y}}{\dot{X}} = \frac{Y(-C+eX)}{X(a-bY)}$ ← this is not defined with $X=0$

rearrange

$\frac{\dot{Y}(a-bY)}{Y} = \frac{\dot{X}(-C+eX)}{X}$

$a\frac{\dot{Y}}{Y} - b\dot{Y} = -C\frac{\dot{X}}{X} + e\dot{X}$

integral both sides

$K + a\ln Y - bY = -C\ln X + eX$

$K = \underbrace{-C\ln X + eX}_{f(x)} \underbrace{- a\ln Y + bY}_{g(y)}$

since K is a constant, this is like a conserved energy of the system

so all trajectories are orbits

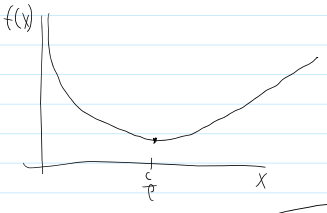
let's analyze $f(x)$

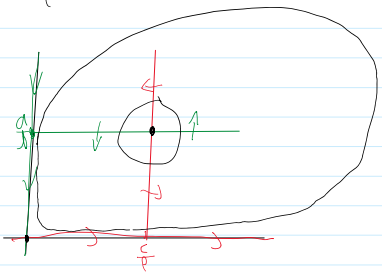
$f'(x) = -\frac{C}{X} + e$

$f'(x) = 0 \Rightarrow X = \frac{C}{e}$ min

$g(y)$ looks the same, its min is $\frac{a}{b}$

so the min K is at $(\frac{c}{e}, \frac{a}{b})$ which is our FP





When we are close to 0, $f(x)$ and $g(y)$ explode, leading to orbits being weird

For a 2×2 linearization, the Jacobian J near a fixed point determines the local behavior.

Let

$$\tau = \text{tr}(J), \qquad \Delta = \det(J).$$

Since

$$\lambda_1 + \lambda_2 = \tau, \qquad \lambda_1\lambda_2 = \Delta,$$

you get:

Determinant Δ	Trace τ	Fixed Point Type
$\Delta < 0$	any	**Saddle** (always unstable)
$\Delta > 0$	$\tau < 0$	Stable node/spiral
$\Delta > 0$	$\tau > 0$	Unstable node/spiral
$\Delta > 0$	$\tau = 0$	Center or special case (need more analysis)
$\Delta = 0$	any	Non-isolated/degenerate (linearization inconclus

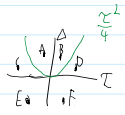
To distinguish **node vs spiral**, look at

$$D = \tau^2 - 4\Delta.$$

D	Eigenvalues	Type
$D > 0$	real, distinct	Node
$D = 0$	repeated real	Improper/star node
$D < 0$	complex conjugates	Spiral (or center if $\tau = 0$)

reminder

$\ln y(t) = \frac{\dot{y}(t)}{y(t)}$



A = stable spiral
B = unstable spiral
C = stable node
D = unstable node
E = F = saddle